

EX. NO: 4 — VIGENERE CIPHER

Aim:

To implement the Vigenere Cipher substitution technique using C.

Algorithm:

1. Read plaintext and key.
2. Repeat the key to match plaintext length.
3. Add plaintext and key values.
4. Apply modulo 26.
5. Display encrypted text.
6. Reverse the operation for decryption.

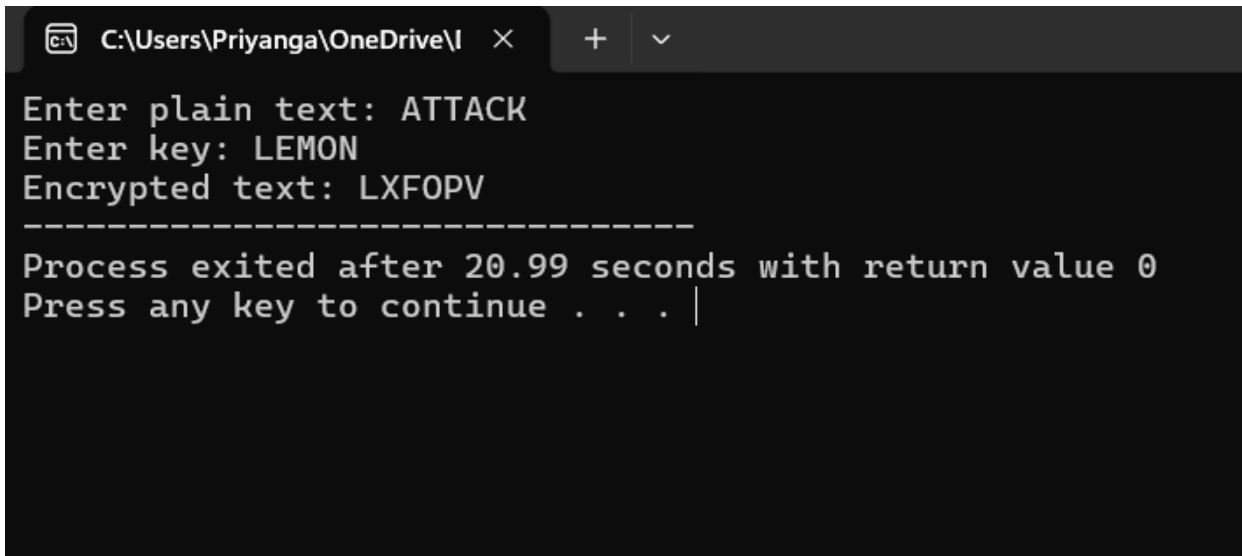
Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
    char text[100], key[50];
    int i, j, len;
    printf("Enter plain text: ");
    scanf("%s", text);
    printf("Enter key: ");
    scanf("%s", key);
    len = strlen(key);
    printf("Encrypted text: ");
    for(i = 0, j = 0; text[i] != '\0'; i++, j++)
    {
        if(j == len)
            j = 0;
```

```
        printf("%c",  
            ((toupper(text[i])-'A') +  
             (toupper(key[j])-'A')) % 26 + 'A');  
    }  
    return 0;  
}
```

Output:



```
C:\Users\Priyanga\OneDrive\I  ×  +  ▾  
Enter plain text: ATTACK  
Enter key: LEMON  
Encrypted text: LXFOPV  
-----  
Process exited after 20.99 seconds with return value 0  
Press any key to continue . . . |
```

Result:

Thus, the Vigenere Cipher substitution technique was implemented successfully using C.